HW 3 — From a Known Score to Langevin Sampling

Mathematical Foundations of Diffusion Models · Monsoon 2026 · IIT Jammu

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Due	Monday 26 October 2026
Marks	40 points — graded on correctness

Individual homework: derivations and programming. Show your working and explain your numerical results. Submit an executed notebook with the mathematical steps, code, labeled figures, and comparison table.

Background: the sampling problem

Suppose we know a probability density but want to generate observations from it using its score. In this homework, the target is the **standard logistic distribution** on the real line:

$$p(x) = \frac{e^{-x}}{(1+e^{-x})^2}, \qquad F(x) = \frac{1}{1+e^{-x}}, \qquad x \in \mathbb{R}.$$

This is a smooth, non-Gaussian density with mean zero and variance $\pi^2/3$. We will derive its score, use it to construct a Langevin sampler, and compare the generated samples with the known target.

Work entirely in **one dimension**. Write

$$s(x) = \frac{d}{dx} \log p(x).$$

The time variable τ denotes elapsed sampling time. The target p stays fixed as τ increases. This differs from diffusion-model time, which indexes a changing family of noisy data distributions.

The distinction between a deterministic velocity and a stochastic drift matters here. We will first examine the ODE $\dot{x} = s(x)$, then use the same score as the drift of an SDE with an additional noise term. You will show why the latter can preserve the target distribution while the former cannot.

How to read this assignment. "Background" and "Given" blocks supply explanations, formulas, or assumptions; they are not additional questions. The lettered **Tasks**, the **Required implementation and experiment settings**, and the **Submission** section specify required work. The marks total 40.

Problem 1 — Derive and interpret the score

(a) Differentiate the log density (4 points)

Starting from $p(x)$, derive $s(x)$ with all intermediate steps. Express your answer both in terms of $F(x)$ and in terms of the hyperbolic tangent.

Given: hyperbolic tangent identity

$$\tanh(u) = \frac{e^u - e^{-u}}{e^u + e^{-u}}.$$

(b) Interpret the vector field (4 points)

Plot $p(x)$ and $s(x)$ on $[-8, 8]$ in separate panels. Determine the sign of $s(x)$ for positive and negative x, locate its zero, and calculate its limits as $x \to \pm\infty$. Explain how the score directs the motion of a particle at each position.
Total: 8 points

Problem 2 — From deterministic motion to a stochastic sampler

Given: dynamics and numerical rules

The deterministic score ODE is

$$\frac{dx}{d\tau} = s(x), \qquad x(0) = x_{\text{init}}.$$

Its explicit Euler update is

$$x_{k+1} = x_k + \eta s(x_k).$$

The Langevin SDE adds Brownian noise:

$$dX_\tau = s(X_\tau)\, d\tau + \sqrt{2}\, dW_\tau.$$

Its **Euler–Maruyama update is provided**:

$$x_{k+1} = x_k + \eta s(x_k) + \sqrt{2\eta}\, z_k, \qquad z_k \overset{\text{iid}}{\sim} \mathcal{N}(0, 1).$$

A Brownian increment over a step of length η has variance η. Multiplication by the SDE coefficient $\sqrt{2}$ therefore gives the numerical noise amplitude $\sqrt{2\eta}$. **Neither numerical rule needs to be derived. No stochastic-calculus derivation is required.**
Background: fixed score, initialization, and noise (no proof required)

The score $s(x) = \partial_x \log p(x)$ has **no explicit dependence on sampling time** τ or step index k. Every Euler–Maruyama step evaluates the same function at the current position x_k. Its value can change because the particle moves; the target density and its score field remain fixed. Changing the numerical step size does not change the target score.

For the logistic target in this homework, $p(x) > 0$ for every $x \in \mathbb{R}$, and its score is smooth, bounded, and globally Lipschitz. Thus the Langevin SDE is well-defined from every finite starting position. We may also draw $X_0 \sim P_{\text{init}}$ from an initial probability distribution on $\mathbb{R}$, independently of the driving Brownian motion. Examples include a Gaussian, a uniform distribution on a finite interval, or a point mass at zero or six.

The initial law need not have the same support as the target. Starting all particles at zero is valid even though the logistic target has support throughout $\mathbb{R}$. Brownian noise allows the particles to spread into regions absent from the initial distribution; the score drift alone would keep particles initialized at zero there. Having full support, however, does not mean having the correct probabilities: reaching the target distribution still requires convergence.

This freedom of initialization is not a claim about every target density. If p vanishes on a region, the expression $\partial_x \log p$ is generally undefined there, so we cannot simply initialize there and expect the drift to repair the problem. Targets confined to a domain may need boundary conditions, such as reflection. Disconnected supports may prevent movement between components. For general targets, the SDE must be well-defined at the starting points, and convergence requires appropriate accessibility and ergodicity conditions. Equality of initial and target supports is neither required in general nor sufficient to establish convergence.

Initialization is separate from the fresh $z_k \sim \mathcal{N}(0, 1)$ used at every numerical step. Each z_k is independent of X_0 and all earlier increments, and therefore of the current state x_k. The initial

law may itself be Gaussian, but its draw and the subsequent Gaussian increments have different roles. The score term is the **drift**; the Brownian term is the **diffusion or noise term**, not a Brownian drift.

For this logistic target, the exact continuous-time Langevin process converges to the target from the stated initializations; you may use this fact without proving it. Initialization affects how long convergence takes. This does not imply exact target samples at a finite running time, and the Euler–Maruyama implementation introduces an additional discretization error. Problem 4 distinguishes these issues from stationarity and finite-sample fluctuations.

(a) Examine deterministic score motion (4 points)

1. Show that every trajectory of the score ODE approaches zero as $\tau \to \infty$. A sign and monotonicity argument is sufficient; a closed-form solution is not required.
2. Explain why this limiting behavior does not reproduce the logistic target density.
3. Plot explicit Euler trajectories from $x_{\text{init}} \in \{-6, -3, -1, 1, 3, 6\}$ with step size 0.02 until $\tau = 20$.

(b) Substitute the score into the Langevin sampler (6 points)

1. Substitute your Problem 1 score into the given Langevin SDE and Euler–Maruyama update. Write the update explicitly in terms of x_k, η, and z_k, and identify its drift and noise terms.
2. For $\eta = 0.02$, explain why the update without noise moves nonzero particles toward zero. State what happens if all particles start exactly at zero.
3. Explain why restoring independent Gaussian increments allows particles to spread from this common initial position, whereas the deterministic score drift alone cannot maintain the logistic target's nonzero variance.

Total: 10 points

Problem 3 — Implement the sampler

Required implementation and experiment settings

Implement the given Euler–Maruyama update, with your explicit logistic score substituted in Problem 2(b), using NumPy or an equivalent numerical library. At each step, evaluate the score at the current particle positions, draw fresh standard Gaussian values, and apply the update. Write this loop yourself; you do not need to derive a numerical solver. No neural network is needed because the score is available analytically.

Use the following experiment:

- Simulate $M = 5{,}000$ independent particles, initially all at zero.
- Compare step sizes $\eta \in \{0.2, 0.05, 0.01\}$.
- Run each experiment to the same elapsed time $\tau_{\max} = 40$, using $K = \tau_{\max}/\eta$ steps.
- Draw fresh, independent Gaussian noise for every particle at every step.
- Retain snapshots at $\tau = 0, 2, 10, 40$ and record your random seed.

Use only the **final position of each independent particle** for the main distribution comparisons.

Background: independent observations

Successive positions along a single trajectory are correlated. Using one final position from each independently simulated particle avoids treating those successive positions as independent observations.

(a) Implement and show the evolving distribution (5 points)

For $\eta = 0.05$, plot the particle histograms at the positive snapshot times and overlay the target density. Describe how the initially concentrated population spreads and approaches the target.

(b) Compare numerical resolutions (5 points)

For each step size, report the final sample mean, variance, and elapsed computational time. Repeat the $\eta = 0.05$ experiment with all particles initially at $x = 6$ and compare the final empirical CDFs from the two initializations. Discuss whether these runs provide evidence that initialization effects have become small.

Total: 10 points

Problem 4 — Explain and check the target distribution

Given: density evolution and comparison formulas

Let $q_\tau(x)$ denote the evolving density of the Langevin particles and $p(x)$ the fixed target. The Fokker–Planck equation is provided:

$$\frac{\partial q_\tau}{\partial \tau} = -\frac{\partial}{\partial x}\big[s(x)q_\tau(x)\big] + \frac{\partial^2 q_\tau(x)}{\partial x^2}.$$

The empirical CDF error is

$$D_M = \sup_x \left|\widehat{F}_M(x) - F(x)\right|.$$

For sorted samples $x_{(1)} \le \cdots \le x_{(M)}$, use the supplied formula

$$D_M = \max_{1\le i\le M}\left\{\frac{i}{M} - F(x_{(i)}),\quad F(x_{(i)}) - \frac{i-1}{M}\right\}.$$

An independent exact logistic observation can be generated by

$$U \sim \text{Uniform}(0,1), \qquad X = \log\frac{U}{1-U}.$$

No derivation of these three formulas is required. The inverse-CDF sampler is supplied only to construct a reference sample, not to replace the Langevin implementation.

Background: what the diagnostics tell us

Density plots are qualitative diagnostics; empirical CDF errors quantify discrepancies. A histogram estimates the generated density, and its appearance depends on sample size and bin choices. D_M compares the empirical CDF with the known target without choosing bins. Here the target plays the role of the data distribution, but there is no training stage because its score is known analytically.

Even exact independent target samples have nonzero D_M due to finite-sample fluctuations. Repeated exact samples give a reference for interpreting whether the Langevin discrepancy is small or large at the same sample size. The range from 20 reference runs is descriptive, not a formal acceptance threshold.

Finite running time, numerical discretization, and finite-sample fluctuations are distinct sources of discrepancy. Smaller step sizes reduce discretization error but need not produce a smaller observed D_M in every experiment. Plots and sample comparisons provide evidence of approximation, not a proof of exact equality for a finite-step sampler.

Stationarity and convergence are different statements — not an assessed proof

For this logistic Langevin SDE, the target density is stationary: if the particles already follow the target distribution, their distribution remains the target as time passes. The score drift

concentrates probability toward the center, while Brownian noise spreads it; at the target density, these effects balance.

Stationarity alone does **not** show that particles initialized from a different distribution converge to the target. Convergence requires a separate argument. For this homework, convergence of the exact logistic Langevin process from the stated initializations is supplied as a fact. **You are not asked to prove convergence, establish ergodicity, or derive conditions guaranteeing either.** The simulations provide a numerical comparison at finite running times, not a convergence proof.

(a) Verify stationarity (6 points)

1. Rewrite the supplied Fokker–Planck equation as $\partial_\tau q_\tau = -\partial_x J_\tau$ and identify the probability flux J_τ.
2. Show that $s(x)p(x) = p'(x)$ and hence that the flux is zero when $q_\tau = p$. Conclude that the target is stationary for the continuous-time Langevin SDE.
3. State the conclusion precisely: if $q_0 = p$, then $q_\tau = p$ for all sampling times $\tau \geq 0$. No convergence argument is required.

(b) Compare generated samples with the target (6 points)

1. **Plot densities.** For each step size, overlay the analytic target density $p(x)$ and a density-normalized histogram of the final generated samples. Label them "Target density" and "Generated samples." Use the same bin edges and axis limits across panels. Describe differences in the center, spread, and tails.
2. **Quantify discrepancies.** Plot each final empirical CDF against F, compute D_M using the supplied formula, and add it to the comparison table from Problem 3.
3. **Construct an exact-sampling reference.** Generate $M = 5{,}000$ independent logistic observations using the supplied inverse-CDF formula. In a reference panel, overlay their density-normalized histogram with p, using the same bins and axes as above. Compute their D_M against F.
4. **Calibrate the error size.** Repeat the exact-sampling experiment 20 times with independent draws. Report the median and range of its D_M values alongside the Langevin errors. State whether the Langevin discrepancies are comparable to ordinary finite-sample fluctuations or noticeably larger.
5. **Interpret the results.** Discuss finite running time, discretization error, and finite-sample fluctuations separately. Explain what your numerical results support and what they do not establish.

Total: 12 points

Submission

Submit one executed Jupyter notebook containing the derivations in LaTeX, code, labeled figures, comparison table, and conclusions. Include package versions and seeds. The notebook should run on a laptop CPU and explain each experiment before presenting its code and output.

Background: scope of the conclusion

Keep the main conclusion precise: **the logistic density is invariant for the continuous-time Langevin SDE; a sufficiently long, sufficiently well-resolved Euler–Maruyama simulation approximates that density.** The deterministic score ODE has a different limiting behavior.